

16 Expected Value

16.1 An Example: GPA

Question: If a student receives 5 A's, 4 B's, and 1 C, what is the student's GPA? (Assume all courses are equally weighted and that an A is worth 4, a B with 3, and a C worth 2.)

1. Explain why $\frac{4+3+2}{3}$ is *not* the correct answer.
2. Show how to correctly calculate the GPA by listing the 10 scores individually.
3. Factor your expression into the following form (fill in the missing numerators):

$$\text{GPA} = 4 \cdot \frac{\quad}{10} + 3 \cdot \frac{\quad}{10} + 2 \cdot \frac{\quad}{10}$$

4. Let X be the random variable that results from randomly selecting a course and recording its grade (on a 4 point scale). Create a probability table for X .
5. How are the numbers in your expression for GPA in #3 related to the numbers in your probability table?

16.2 Generalizing

We can generalize this idea to compute the **mean** (more commonly called **expected value**) of any random variable X . This is denoted either $E(X)$ or μ_X . E is short for expected value. The Greek letter μ (read “mu”) is the Greek version of the letter m for mean. We will use whichever is handier in the moment.

6. Let X be defined by the probability table below. Compute $E(X)$.

value of X	0	1	2	3
probability	0.2	0.3	0.4	0.1

7. Let H be the number of heads in 3 tosses of a fair coin.
 - a. Create a probability table for H
 - b. Compute $E(H)$.
8. Use good mathematical notation to write down the definition:

$$E(X) =$$

9. A raffle has 1000 tickets. Holders of 4 of the tickets get a prize. The other 996 are worth nothing. The four prizes are worth \$500, \$200, \$50, and \$50. Let V be the value of a random raffle ticket.
 - a. Create a probability table for V .
 - b. Compute $E(V)$.
 - c. What does $E(V)$ tell us about the raffle tickets?

10. Let D be the absolute value of the difference (i.e., higher minus lower) between the values of **two 4-sided dice**.
- Create a probability table for D .
 - Compute $E(D)$.
 - What is $p(D = E(D))$?
 - What is $p(D < E(D))$?
 - What is $p(D > E(D))$?

We can do these already now, but there are easier ways that take advantage of properties of expected value that we haven't learned yet.

- In a hand of 5 cards from a standard deck, what is the expected number of diamonds?
- If you roll 5 standard dice, what is the expected number of 6's?

These two are a bit more challenging, but still doable.

- In a hand of 5 cards from a standard deck, what is the expected number of suits?
- If you roll 5 standard dice, what is the expected number of unique numbers rolled?